

→ According to the law of motion

$$\boxed{\frac{d\vec{p}}{dt} = \vec{F}} = -k\vec{x}$$

$$\frac{dp}{dt} = -kx$$

$$\frac{d(mv)}{dt} = -kx$$

$$\boxed{p = mv}$$

→ Momentum (← ~~at~~ dx)

$$m \frac{d}{dt}(v) = -kx$$

$$\boxed{v = \frac{dx}{dt}}$$

→ velocity

$$m \frac{d}{dt} \left(\frac{dx}{dt} \right) = -kx$$

$$m \frac{d^2x}{dt^2} = -kx$$

$$\frac{d^2x}{dt^2} = - \boxed{\frac{k}{m}} x \rightarrow \text{constant}$$

$$\boxed{\omega^2 = \frac{k}{m}}$$

$$\boxed{\frac{d^2x}{dt^2} = -\omega^2 x}$$

→ Diff Eqⁿ Simple Harm Motion. (6H)
= Prove. question



$$E_p = \frac{1}{2} k x^2$$

$$= \frac{1}{2} k (A \cos \omega t)^2$$

$$= \frac{1}{2} k A^2 \cos^2 \omega t$$

$$E_k = \frac{1}{2} m v^2$$

$$= \frac{1}{2} m A^2 \omega^2 \sin^2 \omega t$$

$$= \frac{1}{2} k A^2 \sin^2 \omega t$$

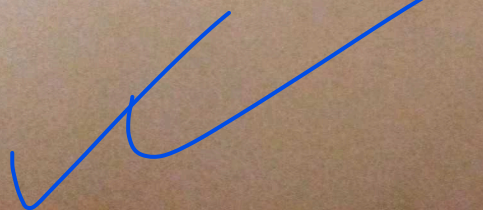
$$E_T = E_p + E_k = \frac{1}{2} k A^2 (\sin^2 \omega t + \cos^2 \omega t)$$

$$= \frac{1}{2} k A^2$$

$$\omega^2 = \frac{k}{m}$$

$$E_T = \frac{1}{2} k A^2$$

↓
(Total energy doesn't depend on Time)
(constant)



$$E_T = E_p + E_k$$

$$\frac{1}{2} k A^2 = \frac{1}{2} k x^2 + \frac{1}{2} m v^2$$

$$\frac{1}{2} m v^2 = \frac{1}{2} k A^2 - \frac{1}{2} k x^2$$

$$m v^2 = k A^2 - k x^2$$

$$m v^2 = k (A^2 - x^2)$$

$$v^2 = \frac{k}{m} (A^2 - x^2)$$

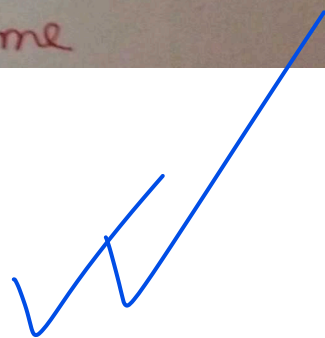
Exm
Prove

$$v = \omega \sqrt{A^2 - x^2}$$

$$\omega^2 = \frac{k}{m}$$

$$\omega = \sqrt{\frac{k}{m}}$$

→ No need time



$$F_{\text{restoring}} = -mg \sin \theta$$

$$F = -mg \theta$$

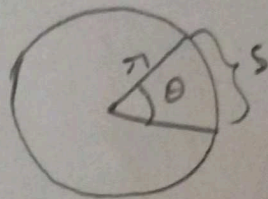
if, θ is very small,
 $\sin \theta \approx \theta$

$$x = L \theta$$

$$\frac{dx}{dt} = L \frac{d\theta}{dt}$$

$$a = L \frac{d^2 \theta}{dt^2}$$

θ = angular displacement
 x = linear displacement



We know,

$$F = -mg \sin \theta$$

$$\Rightarrow ma = -mg \sin \theta$$

$$\Rightarrow a = -g \theta$$

$$\Rightarrow L \frac{d^2 \theta}{dt^2} = -g \theta$$

$$\Rightarrow \frac{d^2 \theta}{dt^2} = -\frac{g}{L} \theta$$

$$\Rightarrow \frac{d^2 \theta}{dt^2} = -\omega^2 \theta$$

$$\Rightarrow \frac{d^2 \theta}{dt^2} + \omega^2 \theta = 0 \rightarrow \text{For simple pendulum}$$

now, $\omega = \frac{g}{L}$

$$\omega = \sqrt{\frac{g}{L}}$$

$$\Rightarrow \frac{2\pi}{T} = \sqrt{\frac{g}{L}}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{L}{g}}$$

[proved]

$$\text{in spring, } \omega = \sqrt{\frac{k}{m}}$$

$$\text{in angle, } \omega = \sqrt{\frac{g}{L}}$$

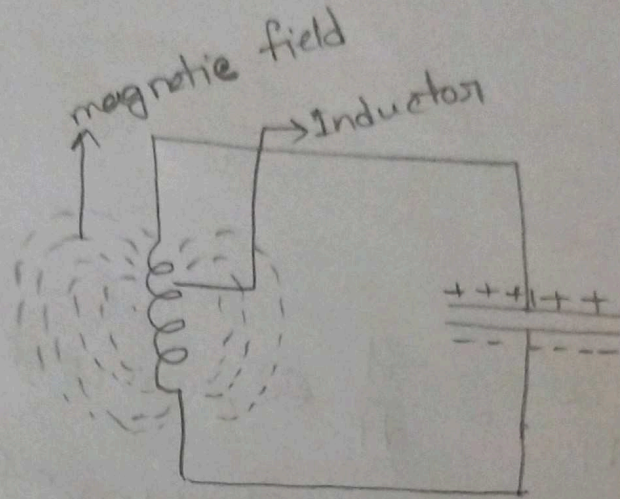
$$\text{angular displacement} \\ \frac{d^2 \theta}{dt^2} + \omega^2 \theta = 0$$

$$\text{linear displacement} \\ \frac{d^2 x}{dt^2} + \omega^2 x = 0$$

LC-Circuit

$$V_L = L \frac{di}{dt}$$

L = inductance
 i = current flow



$$Q = CV_C$$

$$V_C = \frac{Q}{C}$$

According to Kirchhoff's law,

$$V_L + V_C = 0$$

$$\Rightarrow L \frac{di}{dt} + \frac{Q}{C} = 0 \quad i = \frac{dQ}{dt}$$

$$\Rightarrow L \frac{d}{dt} \left(\frac{dQ}{dt} \right) + \frac{Q}{C} = 0$$

$$\Rightarrow L \frac{d^2 Q}{dt^2} + \frac{Q}{C} = 0$$

$$\frac{d^2 Q}{dt^2} + \frac{1}{LC} Q = 0$$

$$\frac{d^2 Q}{dt^2} + \omega_0^2 Q = 0$$

$$\omega_0^2 = \frac{1}{LC}$$

$Q = Q_0 \sin \omega t \rightarrow$ reverse direction to time count

$Q = Q_0 \cos \omega t$ [most suitable]

Damped Harmonic Motion

Lecture: 9

①

Frictional Force:

$$\vec{F}' \propto -v$$

$$F' = -bv$$

For horizontal force:

$$ma = -kx - bv$$

$$m \frac{dx}{dt} = -kx - b \frac{dx}{dt}$$

$$\frac{dx}{dt} = -\frac{k}{m}x - \frac{b}{m} \frac{dx}{dt}$$

$$\frac{dx}{dt} = -\omega^2 x - \gamma \frac{dx}{dt}$$

$$\omega^2 = \frac{k}{m}$$

damping constant

$$\gamma = \frac{b}{m}$$

$$\frac{dx}{dt} + \gamma \frac{dx}{dt} + \omega^2 x = 0$$

→ equation of Damped Harmonic motion. (D.H.M)

→ For this part it is a D.H.M.

Trial Solution:

$$\frac{d^2x}{dt^2} + \gamma \frac{dx}{dt} + \omega_0^2 x = 0$$

$$\text{Let, } x = B e^{pt} \quad B = \text{Amplitude}$$

$$\frac{dx}{dt} = B p e^{pt}$$

$$\frac{d^2x}{dt^2} = B p^2 e^{pt}$$

$$B p^2 e^{pt} + \gamma B p e^{pt} + \omega_0^2 B e^{pt}$$

$$p^2 + \gamma p + \omega_0^2 = 0 \quad \text{--- (1)}$$

Comparing with $ax^2 + bx + c = 0$, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

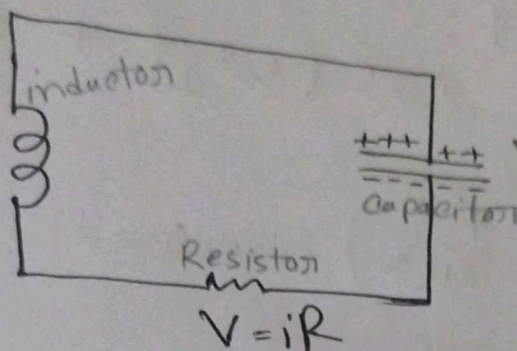
$$\text{So, } p = \frac{-\gamma \pm \sqrt{\gamma^2 - 4\omega_0^2}}{2}$$

$$= -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2 - 4\omega_0^2}{4}}$$

$$p = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega_0^2}$$

RLC Circuit

$$V_L = L \frac{di}{dt}$$



$$V_C = \frac{Q}{C}$$

- * Some energy will be captured by resistor.
- * Discharged to charged.

According to KVL:
 $V = V_L + V_C$

→ (absorb energy each cycle)

$$\Rightarrow iR + L \frac{di}{dt} + \frac{Q}{C} = 0$$

$$\Rightarrow L \frac{d^2Q}{dt^2} + R \frac{dQ}{dt} + \frac{1}{C} Q = 0$$

$$\Rightarrow \frac{d^2Q}{dt^2} + \frac{R}{L} \frac{dQ}{dt} + \frac{1}{LC} Q = 0$$

$$\left[\begin{array}{l} \gamma = \frac{R}{L} \\ \omega_0^2 = \frac{1}{LC} \end{array} \right]$$

$$\frac{d^2Q}{dt^2} + \gamma \frac{dQ}{dt} + \omega_0^2 Q = 0$$